

Leo Patrick Mulholland

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RESEARCH STATEMENT

I am interested in the signature of heavy atomic elements in spectroscopy. I am highly proficient in the computation of atomic structures, radiative rates, collision strengths, photoionisation cross sections and recombination rate coefficients for heavy elements, as well as collisional radiative modelling. This data is being produced for the modelling of kilonovae and other r -process sites, as well as experimental corroboration in laboratory plasmas. Alongside continuing extensive atomic calculations, I am additionally eager to expand into an interdisciplinary skill set by integrating the growing atomic data sets into large-scale radiative transfer modelling, in an effort to improve kilonova simulation and aid observational interpretation.

EDUCATION AND RESEARCH

Oct 2023 - present	PhD (Physics) <i>Theoretical modelling of neutron star mergers:</i> <i>Atomic Theory and Radiation Transport</i> Supervisors: C. Ballance, S. Sim and C. Ramsbottom	Queen's University Belfast
Jun 2022 - Sep 2022	Research intern (Physics)	University of Oldenburg
Sep 2019 - Jul 2023	MSci (Theoretical Physics, first class honours)	Queen's University Belfast

TEACHING AND ACADEMIC SERVICES

2024 - 2025	Supervision of the MSci thesis of Beth Campbell. This work has contributed directly to an in-preparation article in kilonova science.
2024 - 2025	Member of organising committee for astrophysics seminar series at QUB.
2023 - present	Participation in outreach programmes such as 'Girls in Maths & Physics' to aid diversity, equity and inclusion efforts within QUB.
2023 - present	Member of two organising committees for HEAVYMETAL workshops. Including arrangement of excursions and meals, discussion of scientific programmes and venues. I also acted as session chair on three occasions.
2023 - present	Teaching assistant and tutor for classical mechanics (level 2), real analysis and linear algebra (level 1). These included lone delivery of tutorials, supervision of computer classes and management of an online help desk.
2022 - present	Homework grading for undergraduate mathematics courses.

TECHNICAL SKILLS

I am experienced in the use of general atomic codes. During my PhD, I have published open-access collisional data using the DARC and Breit-Pauli R-matrix codes, as well as the AUTOSTRUCTURE code. This required a working knowledge of high-performance computing and data analysis.

I am adept in FORTRAN and PYTHON programming. I have publicly available open-source codes for the analysis and streamlining of atomic computations. I have directly contributed to the maintenance and improvement of FORTRAN production-level atomic codes such as those in the DARC suite. On my GitHub I host open-source FORTRAN codes for processing resonant excitation calculations as well as PYTHON codes for modelling spectra, calculating charge state balance with fast electrons, and various other utilities for open science. I have experience in the use of the Monte-Carlo Radiative Transfer code TARDIS. I am also proficient in the memory-safe language RUST. Versed in C/C++, with an accreditation from Codecademy. Skilled in the use of OpenMP/MPI, as well as GPU acceleration - for which I have an accreditation from NVIDIA.

PRESENTATIONS

September 2026	Invited talk at BRIDGCE, University of Hertfordshire, England. (upcoming)
January 2026	Contributed talk at AtomicRadTran, LANL, USA.
June 2025	Poster and flash talk at EAS, University College Cork, Ireland.
May 2025	Seminar at Auburn University, USA.
July 2024	Contributed talk at BRIDGCE, University of Surrey, England.
September 2024 - present	Seminars in the internal ARC series at QUB, Northern Ireland.
September 2023 - present	Contributed talks at all HEAVYMETAL workshops.

PUBLICATIONS

Six peer-reviewed open-access articles, five of which as first author (additional two in prep.). **Five** of these six are in kilonova science with a specialisation in atomic data production and modelling (~ 40 citations). Two further works as co-author are in review.

First Author Works

L. P. Mulholland et al. *Electron impact excitation of Te IV and V and level resolved R-matrix photoionization of Te I – IV with application to modelling of AT2017gfo*. MNRAS **546** (advance access) (2026).

L. P. Mulholland et al. *On the use of the Axelrod formula for thermal electron collisions in astrophysical modelling*. JQSRT **345** 109545 (2025).

L. P. Mulholland et al. *Collisional and radiative data for tellurium ions in kilonovae modelling and laboratory benchmarks*. MNRAS **534** 3423–3438 (2024).

L. P. Mulholland et al. *New radiative and collisional atomic data for Sr II and Y II with application to kilonova modelling*. MNRAS **532** 2289–2308 (2024).

L. P. Mulholland et al. *Non-analytic behaviour in large-deviations of the susceptible-infected-recovered model under the influence of lockdowns*. New J. Phys. **25** 13034 (2023).

Coauthored works

M. McCann, **L. P. Mulholland** et al. . *Luminosity predictions for the first three ionization stages of W, Pt, and Au to probe potential sources of emission in kilonova*. MNRAS **538** 537–552 (2025).

S. A. Sim et al. (**inc L. P. Mulholland**). *Late-time emission-line profiles from kilonova models*. MNRAS **551** (advance access) (2026).

N. Ferguson, **L. P. Mulholland** et al.. *Excitation and Recombination Data for Nd I–V with Applications to Kilonovae*. MNRAS, accepted.

In preparation

L. P. Mulholland et al. *Ce II–IV emission in kilonova with R-matrix collision strengths and distorted wave recombination rates*. In preparation.

L. P. Mulholland et al. *Nebular kilonova emission: implications of first r-process peak abundances from atomic and nuclear calculations*. In preparation.

GRANTS AND AWARDS

2026	O’Sullivan Masterclass. Royal Irish Academy. Awarded every year to one student from each institution on the island of Ireland.
2025 - 2027	STFC computing grant on DiRAC. Led proposal and science case. 4.21 Mcpuh, 0.49 Mgpuh awarded over two years.
2023	Earnshaw Award. Awarded to the top undergraduate thesis each year by the Institute of Physics (Ireland).
2019 - 2023	Four undergraduate prizes awarded by QUB, including the Bates prize.

REFERENCES

Prof. Catherine Ramsbottom

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Prof. Stuart Loch

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